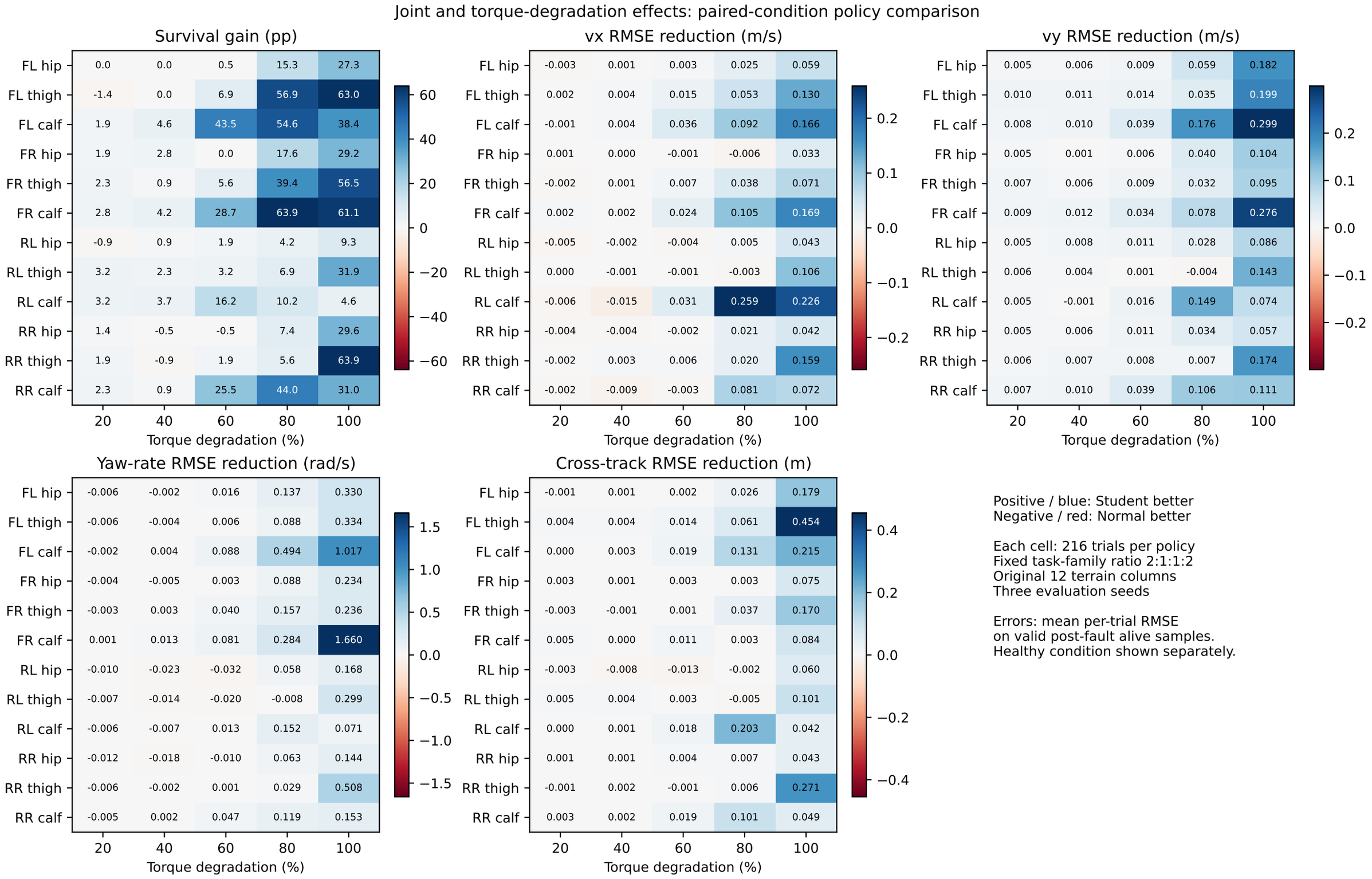
Joint and torque-degradation effects: paired-condition policy comparison
Survival gain (pp)
FL hip 0.0 0.0 0.5 15.3 27.3
FL thigh -1.4 0.0 6.9 56.9 63.0
FL calf 1.9 4.6 43.5 54.6 38.4
FR hip 1.9 2.8 0.0 17.6 29.2
FR thigh 2.3 0.9 5.6 39.4 56.5
FR calf 2.8 4.2 28.7 63.9 61.1
RL hip -0.9 0.9 1.9 4.2 9.3
RL thigh 3.2 2.3 3.2 6.9 31.9
RL calf 3.2 3.7 16.2 10.2 4.6
RR hip 1.4 -0.5 -0.5 7.4 29.6
RR thigh 1.9 -0.9 1.9 5.6 63.9
RR calf 2.3 0.9 25.5 44.0 31.0
60 40 20 0 −20 −40 −60
20 40 60 80 100
Torque degradation (%)
vx RMSE reduction (m/s)
FL hip -0.003 0.001 0.003 0.025 0.059
FL thigh 0.002 0.004 0.015 0.053 0.130
FL calf -0.001 0.004 0.036 0.092 0.166
FR hip 0.001 0.000 -0.001 -0.006 0.033
FR thigh -0.002 0.001 0.007 0.038 0.071
FR calf 0.002 0.002 0.024 0.105 0.169
RL hip -0.005 -0.002 -0.004 0.005 0.043
RL thigh 0.000 -0.001 -0.001 -0.003 0.106
RL calf -0.006 -0.015 0.031 0.259 0.226
RR hip -0.004 -0.004 -0.002 0.021 0.042
RR thigh -0.002 0.003 0.006 0.020 0.159
RR calf -0.002 -0.009 -0.003 0.081 0.072
0.2 0.1 0.0 −0.1 −0.2
20 40 60 80 100
Torque degradation (%)
vy RMSE reduction (m/s)
FL hip 0.005 0.006 0.009 0.059 0.182
FL thigh 0.010 0.011 0.014 0.035 0.199
FL calf 0.008 0.010 0.039 0.176 0.299
FR hip 0.005 0.001 0.006 0.040 0.104
FR thigh 0.007 0.006 0.009 0.032 0.095
FR calf 0.009 0.012 0.034 0.078 0.276
RL hip 0.005 0.008 0.011 0.028 0.086
RL thigh 0.006 0.004 0.001 -0.004 0.143
RL calf 0.005 -0.001 0.016 0.149 0.074
RR hip 0.005 0.006 0.011 0.034 0.057
RR thigh 0.006 0.007 0.008 0.007 0.174
RR calf 0.007 0.010 0.039 0.106 0.111
0.2 0.1 0.0 −0.1 −0.2
20 40 60 80 100
Torque degradation (%)
Yaw-rate RMSE reduction (rad/s)
FL hip -0.006 -0.002 0.016 0.137 0.330
FL thigh -0.006 -0.004 0.006 0.088 0.334
FL calf -0.002 0.004 0.088 0.494 1.017
FR hip -0.004 -0.005 0.003 0.088 0.234
FR thigh -0.003 0.003 0.040 0.157 0.236
FR calf 0.001 0.013 0.081 0.284 1.660
RL hip -0.010 -0.023 -0.032 0.058 0.168
RL thigh -0.007 -0.014 -0.020 -0.008 0.299
RL calf -0.006 -0.007 0.013 0.152 0.071
RR hip -0.012 -0.018 -0.010 0.063 0.144
RR thigh -0.006 -0.002 0.001 0.029 0.508
RR calf -0.005 0.002 0.047 0.119 0.153
1.5 1.0 0.5 0.0 −0.5 −1.0 −1.5
20 40 60 80 100
Torque degradation (%)
Cross-track RMSE reduction (m)
FL hip -0.001 0.001 0.002 0.026 0.179
FL thigh 0.004 0.004 0.014 0.061 0.454
FL calf 0.000 0.003 0.019 0.131 0.215
FR hip 0.003 0.001 0.003 0.003 0.075
FR thigh -0.003 -0.001 0.001 0.037 0.170
FR calf 0.005 0.000 0.011 0.003 0.084
RL hip -0.003 -0.008 -0.013 -0.002 0.060
RL thigh 0.005 0.004 0.003 -0.005 0.101
RL calf 0.000 0.001 0.018 0.203 0.042
RR hip 0.001 0.001 0.004 0.007 0.043
RR thigh -0.001 0.002 -0.001 0.006 0.271
RR calf 0.003 0.002 0.019 0.101 0.049
0.4 0.2 0.0 −0.2 −0.4
20 40 60 80 100
Torque degradation (%)
Positive / blue: Student better
Negative / red: Normal better
Each cell: 216 trials per policy
Fixed task-family ratio 2:1:1:2
Original 12 terrain columns
Three evaluation seeds
Errors: mean per-trial RMSE
on valid post-fault alive samples.
Healthy condition shown separately.